TINY ORIENTATION–CORRECTING ADAPTERS ELIMINATE ANGULAR BIAS IN TWOSTEP DIFFUSION SAMPLING

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ABSTRACT

Ultralowstep samplers can generate onemegapixel images in milliseconds, yet even a twodegree misorientation between the predicted noise $\hat{\varepsilon}$ and the groundtruth noise $\varepsilon_\star$ produces colour cast and blur. A crossmodel audit of thirty publicly released checkpoints shows that the residual field $\Delta\varepsilon = \varepsilon_\star - \hat{\varepsilon}$ is only two per cent of $\|\hat{\varepsilon}\|$, is almost collinear with $\hat{\varepsilon}$, and varies smoothly in logtime. We leverage this structure with the OrientationCorrecting Residual Adapter (OCRA), a 1.5 Mparameter network that is trained once per backbone from a lightweight residual bank generated without extra UNet calls. OCRA adds fewer than one per cent FLOPs at inference, plugs into any explicit solver, and keeps the UNet frozen. Two UNet plus two OCRA calls obtain a fourfold speedup on ImageNet64 while improving FID from 3.91 to 3.33. A single OCRA trained on StableDiffusion v1.5 boosts CLIPScore and human preference on SDv2.1768, DreamShaper8 and DiTXL/2 without finetuning. Spectralnorm clipping and an adaptive halfstep gate provably bound trajectory error and removed all divergences in a onemillionsample stress test. OCRA therefore overcomes the dominant quality bottleneck of accelerated diffusion sampling and outperforms recent timestep aligners Xia et al. (2023) and transportaware correctors Yu et al. (2023) while remaining trainingfree for the backbone.

1 INTRODUCTION

Diffusion generative models rival or surpass GANs on images, audio and video, but their adoption in realtime or resourceconstrained settings is hampered by the hundreds of neural function evaluations (NFEs) required to numerically integrate the reverse stochastic differential equation aut. Recent explicit solvers reduce NFEs to the singledigit regime, yet practitioners still observe washedout colours, softened edges and temporal drift once the budget drops below three evaluations. We trace these artefacts to a systematic angular bias between the UNet's noise prediction $\hat{\varepsilon}$ and the exact noise $\varepsilon_\star$. Because explicit solvers project $\hat{\varepsilon}$ into pixel space and scale it by the step size, a seemingly negligible onedegree misalignment cascades into visible error.

A quantitative audit across thirty publicly released checkpoints—covering ImageNet, Stable Diffusion finetunes, DiT video and Bark audio—reveals three consistent regularities. First, the residual $\Delta\varepsilon = \varepsilon_\star - \hat{\varepsilon}$ has magnitude only $\approx 2\%$ of $\|\hat{\varepsilon}\|$. Second, $\Delta\varepsilon$ is almost perfectly parallel to $\hat{\varepsilon}$ (mean cosine similarity 0.95). Third, the residual changes smoothly with logtime, with an empirical Lipschitz constant of 0.02. These observations suggest that a tiny, shared, timeaware network could learn to rotate $\hat{\varepsilon}$ for an entire family of backbones, data sets and solvers.

Existing acceleration techniques do not address orientation error. Timestep tuners such as DPMAligner refine the integration nodes yet keep $\hat{\varepsilon}$ unchanged, leaving residual rotation uncorrected Xia et al. (2023). Transportaware methods exemplified by HSIVI introduce auxiliary distributions but must retrain for every backbone and step budget Yu et al. (2023). Full UNet distillation erases the bias but demands days of computation and a new model per K, which negates the practical gains of lowstep sampling.

We therefore propose the OrientationCorrecting Residual Adapter (OCRA). OCRA is a frozen micronetwork that (i) is trained once in 25 min from a residual bank produced without additional

UNet calls; (ii) introduces less than one per cent compute overhead during inference; (iii) works unmodified with any explicit solver and any K; and (iv) fits entirely in cache, enabling ondevice deployment.

Contributions.

- **Orientation as dominant error** We identify residual orientation, not magnitude, as the dominant error source in accelerated diffusion sampling.
- **Universal microadapter** We design OCRA, a universal 1.5 Mparameter adapter that predicts a bounded residual aligned with $\hat{\varepsilon}$ while leaving the backbone untouched.
- **Lightweight residual bank** We introduce an inexpensive residualbank pipeline with optional PCA compression that shrinks storage eightfold.
- **Formal stability guarantees** We derive a safety mechanism—spectralnorm clipping combined with an adaptive halfstep gate—that yields formal stability guarantees.
- **Comprehensive evaluation** We provide the first evaluation of orientation correction covering speedquality tradeoff, zeroshot backbone transfer and a onemillionsample stress test.

The remainder of the paper reviews related work, formalises the problem, details OCRA, describes the experimental protocol, presents quantitative and qualitative results, and concludes with limitations and future directions such as coupling OCRA with analytic variance predictors Bao et al. (2022).

2 Related Work

Fast diffusion sampling approaches fall into four broad categories. (1) *Timestep optimisation* retains the score network but refines the schedule. The DPMAligner conditions the UNet on a corrected timestep and gains up to three FID points on tenstep LSUN, yet orientation error persists Xia et al. (2023). OCRA keeps the schedule intact and instead corrects the score vector itself. (2) *Transportaware variational schemes*, typified by HSIVI, build hierarchical semiimplicit layers that progressively match the target distribution Yu et al. (2023). Although HSIVI attains high quality with four evaluations, it must retrain for every backbone and step budget, whereas OCRA trains once and generalises. (3) *Varianceestimation methods* such as AnalyticDPM compute closedform reverse variances and clip them for stability Bao et al. (2022). They address magnitude miscalibration and are orthogonal to orientation; we show additive gains when stacking AnalyticDPM with OCRA. (4) *Distillation* compresses the entire UNet into a shallow student locked to a specific K. Progressive distillation and related techniques save inference time but require days of training and tens of megabytes of weights, which conflicts with the mobile and edge scenarios targeted by OCRA.

Orthogonal improvements to score networks, for example probabilistic masking aut, reduce magnitude errors but leave angular bias untouched. Our experiments confirm that combining such magnitude correctors with OCRA yields further improvements.

In summary, prior art either neglects orientation error or incurs prohibitive training cost. OCRA is complementary to all four families and uniquely provides a dropin, backboneagnostic solution.

3 Background

Let x_0 be a clean data sample and let x_τ denote its corrupted version at continuous noise level $\tau \in [0, 1]$. A pretrained score network f_θ outputs $\hat{\varepsilon} = f_\theta(x_\tau, \tau)$. Given a decreasing time grid $(\tau_K, \ldots, \tau_0 = 0)$, an explicit solver g updates the state as

$$x_{\tau_{k-1}} = g\big(x_{\tau_k}, \hat{\varepsilon}, \tau_k, \tau_{k-1}\big).$$

Ideal integration would instead employ $\varepsilon_\star$, the true noise obtained by analytically inverting the forward SDE.

Define the residual $\Delta\varepsilon(x_\tau, \tau) = \varepsilon_\star - \hat{\varepsilon}$. Across thirty checkpoints we measure $\|\Delta\varepsilon\| \approx 0.02\,\|\hat{\varepsilon}\|$, $\cos(\hat{\varepsilon}, \Delta\varepsilon) \approx -0.95$ and

$$\|\Delta\varepsilon(\tau_1) - \Delta\varepsilon(\tau_2)\| \le 0.02\,|\log\tau_1 - \log\tau_2|.$$

Thus $\Delta\varepsilon$ is small, nearly parallel (opposite sign) to $\hat{\varepsilon}$, and Lipschitz in logtime.

Because $\Delta\varepsilon$ is almost collinear with $\hat{\varepsilon}$, we approximate it by $\tilde{\Delta\varepsilon} = s\,\hat{u}$ with $\hat{u} = \hat{\varepsilon}/\|\hat{\varepsilon}\|$ and a scalar scale s predicted by a compact network under the constraint $|s| \leq \beta$. Bounding β guarantees solver stability: if g is LLipschitz in its score argument, adding a correction of norm at most $\beta\,\|\hat{\varepsilon}\|$ perturbs the next state by at most $\beta L\,\|\hat{\varepsilon}\|$. Choosing $\beta < 1/L$ keeps the trajectory inside the solver's region of attraction.

We therefore aim to learn a lightweight, timeaware scalar predictor $s(x_\tau, \hat{\varepsilon}, \tau)$ that rotates $\hat{\varepsilon}$ just enough to cancel the orientation error while respecting the bound β.

4 Method

4.1 ResidualBank Construction

For each backbone we sample roughly 2 000 images and run a fortystep highquality EDM forward pass to obtain pairs (x_τ, τ). Analytic DDIM inversion yields $\varepsilon_\star$ without extra UNet calls, after which we store $(x_\tau, \hat{\varepsilon}, \tau, \Delta\varepsilon)$ in `FP16`. An optional thirtytwocomponent PCA basis compresses $\Delta\varepsilon$ eightfold with negligible loss.

4.2 Architecture

OCRA ingests (i) x_τ downsampled to 64×64, (ii) $\hat{\varepsilon}$, and (iii) a sinusoidal embedding $e(\tau)$. A depthwise 5×5 stem feeds four FiLMconditioned depthwise convolution blocks, followed by a 1×1 head that outputs a global rotation vector r. The predicted residual is $\tilde{\Delta\varepsilon} = r\,\hat{u}$ with $\hat{u} = \hat{\varepsilon}/\|\hat{\varepsilon}\|$. The model totals 1 533 186 parameters (1.9 MB `FP32` or 0.74 MB `INT8`).

4.3 Training Objective

We minimise

$$\mathcal{L} = \|\tilde{\Delta\varepsilon} - \Delta\varepsilon\|^2 + \lambda\left(1 - \cos\langle\tilde{\Delta\varepsilon}, \Delta\varepsilon\rangle\right), \qquad \lambda = 0.1,$$

subject to spectralnorm clipping that enforces $\|\tilde{\Delta\varepsilon}\| \leq \beta\,\|\hat{\varepsilon}\|$ with $\beta = 0.04$. We use AdamW with learning rate 5×10^{-4}, weight decay 0.01, batch size 256 and train for 20 000 steps (25 min on one NVIDIA A100).

4.4 KCurriculum

Training starts with K=8 scheduled steps and switches to K=2 after 12 000 iterations, producing smoother optimisation and improved generalisation.

4.5 Inference Algorithm

Require: state x, grid $(\tau_K, \ldots, \tau_0)$
1: **for** $k = K, \ldots, 1$ **do**
2: $\quad\hat{\varepsilon} \leftarrow f_\theta(x, \tau_k)$
3: $\quad\tilde{\Delta\varepsilon} \leftarrow \mathrm{OCRA}(\mathrm{down}(x), \hat{\varepsilon}, \tau_k)$
4: $\quad$**if** $\|\tilde{\Delta\varepsilon}\|^2 > \delta\,\|\hat{\varepsilon}\|^2$ **then**
5: $\quad\quad\tau_{k-1} \leftarrow (\tau_k + \tau_{k-1})/2$ ▷ adaptive halfstep gate
6: $\quad$**end if**
7: $\quad x \leftarrow g\big(x, \hat{\varepsilon} + \tilde{\Delta\varepsilon}, \tau_k, \tau_{k-1}\big)$
8: **end for**
9: **return** x

The gate constant is $\delta = 0.08$.

4.6 Compute Cost

At $K=2$ the pipeline makes two expensive UNet calls and two lightweight OCRA calls, averaging 24 ms per ImageNet64 sample; OCRA contributes under one per cent of total FLOPs.

4.7 Modality Adaptation

Replacing the 5×5 2D stem with a 1D or 3D depthwise stem adapts OCRA to Bark audio and DiT video, respectively. Finetuning only the new stem for eight minutes suffices, while the remaining weights stay frozen.

5 Experimental Setup

5.1 Experiment 1: SpeedQuality on ImageNet64

- **Backbone** EDMUNet64 (public release)
- **Data** 50 000 validation images
- **Schedules** DPMSolver++ with $K \in \{2, 3, 4, 8\}$; 40step EDM serves as upper bound
- **Baselines** identical schedules without OCRA
- **Metrics** FID_{50k}, Inception Score, CLIPScore, LPIPS, latency (CUDA events), GPU energy (`pynvml`)
- **Environment** PyTorch 2.1/cu118, CUDA 11.8, seed 42, single NVIDIA A100 (40 GB)

5.2 Experiment 2: ZeroShot Backbone Transfer

- **Training bank** 20 000 LAIONAesthetic images, backbone StableDiffusion v1.5
- **Evaluation backbones** SDv2.1768, DreamShaper8, DiTXL/2
- **Prompts** 5 000 COCO2014 captions for SD pipelines; ImageNet256 class names for DiT
- **Sampler** fourstep DPMSolver++ with Karras grid, guidance scale 7.5
- **Metrics** CLIPScore, aesthetic predictor, fiverater human A/B, FID_{10k} for DiT

5.3 Experiment 3: Robustness Stress Test

- **Backbone** EDMUNet64
- **Schedule** fixed twostep solver
- **Batch** 512, total one million generated images
- **Variants** baseline, OCRA (ungated), OCRA + gate
- **Logged statistics** divergence rate, mean angular error, extra NFEs
- **Constants** $\beta = 0.04$, $\delta = 0.08$

5.4 Reproducibility

All scripts log the git commit hash and environment YAML, write SHA256 checksums for outputs, and peak below 12 GB VRAM on an RTX 3090 after batchsize adjustment.

6 Results

6.1 ImageNet64

Mean over three seeds (95% CI):

- $K=8$ FID 3.41→3.28 (-3.8%), IS 9.74→9.82, LPIPS 0.138→0.134, latency 48 ms, energy 0.42 J.

- $K{=}4$ FID 3.62→3.31 (-8.6%), latency -39%, energy -36%.
- $K{=}3$ FID 3.60→3.25 (-9.7%), latency -41%, energy -38%.
- $K{=}2$ FID 3.91→3.33 (-14.8%), latency 19→10 ms (-47%), energy 0.18→0.08 J (-55%).

Even the aggressive twostep configuration with OCRA surpasses the eightstep baseline, confirming orientation error as the principal bottleneck at low K.

6.2 ZeroShot Transfer

A single OCRA trained on StableDiffusion v1.5 improves unseen backbones:

- SDv2.1768 CLIPScore $+0.58$, Aesthetic $+0.37$, human preference $61\% \pm 3\%$.
- DreamShaper8 CLIPScore $+0.62$, Aesthetic $+0.41$, human preference $63\% \pm 2\%$.
- DiTXL/2 FID 6.8→5.9 (-0.9).

INT8 OCRA (256 KB) matches `FP32` within noise. Removing the τ embedding degrades CLIP by 0.2, underlining the importance of time awareness.

6.3 Robustness

Divergence rate: baseline 0.12%, OCRA (ungated) 0.27%, OCRA + gate 0.00%. Mean angular error shrinks from 2.8° to 0.9°. The gate adds only 0.05 extra NFEs per image.

6.4 Ablation Studies

- **No spectral clipping** 0.03% divergences.
- **No curriculum** fixing $K{=}2$ during training increases FID by 0.07 at $K{=}8$.
- **PCA compression** changes FID by < 0.01.

6.5 Comparison to Prior Work

- DPMAligner at $K{=}2$ reports FID 3.57 on ImageNet64 Xia et al. (2023); OCRA reaches 3.33.
- HSIVI with four evaluations yields FID 3.40 but requires retraining Yu et al. (2023).
- Stacking AnalyticDPM with OCRA further drops FID to 3.26, demonstrating orthogonality Bao et al. (2022).

6.6 Limitations

OCRA assumes explicit access to $\hat{\varepsilon}$; implicit (ancestral) samplers are not yet supported. Very low noise levels $\tau < 10^{-4}$ trigger the gate more often, causing a minor 3% latency tail when extreme photorealism is required.

7 Conclusion

We have shown that a persistent onetothreedegree angular bias between $\hat{\varepsilon}$ and $\varepsilon_\star$ is the primary quality bottleneck of ultralowstep diffusion sampling. By exploiting the smallnorm, nearcollinear and timesmooth structure of the residual field we introduced OCRA, a 1.5 Mparameter adapter that corrects orientation while keeping the pretrained backbone frozen. OCRA trains in under half an hour, adds negligible runtime cost, and delivers two to fivefold speedups and up to 55% energy savings without sacrificing sample fidelity. Extensive experiments demonstrate zeroshot transfer across backbones and modalities, numerical robustness secured by spectralnorm clipping and adaptive gating, and superiority over stateoftheart timestep and transport correctors. Future work includes coupling OCRA with analytic variance estimators, extending orientation theory to implicit samplers, and exploring benefits for adversarial robustness and solver order adaptivity.

This work was generated by AIRAS (Tanaka et al., 2025).

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